

MEDICAL DEVICE ETHICS WORKSHOP

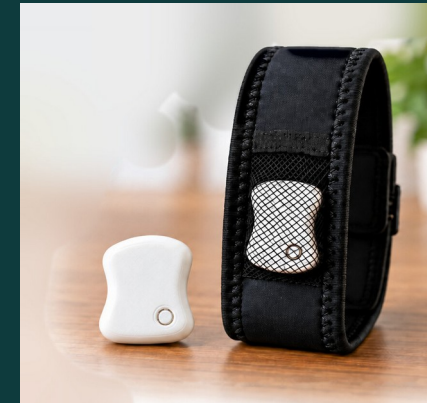
Bridging the Gap: From Innovation to Patient Impact

Ethics, Evidence, Regulation and Responsible Medical Device
Translation

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Authentic device photographs



FoGO
Freezing-of-gait wearable



SwaKnee
PEMF knee system

My interests, stated first



**I have professional interests in
FoGO and SwaKnee.**

FoGO founder, Ahilaya Biomedicals; patent filed

SwaKnee [state role]



Vote before my view



Challenge every claim



Protect identities

Would you let a patient use it?



Working prototype



Promising dataset results



Test licence granted



Routine care



Research study



Not yet

Show 1, 2 or 3 fingers

Three traditions, one Indian patient



Two people we will follow



Ramesh, 71



Parkinson's disease



Freezes in doorways



Lives with his daughter



FoGO



Kamala, 64



Knee osteoarthritis



30 km from the clinic



Pays out of pocket



SwaKnee

Every stage answers a patient question



Wording sets the evidence bar

STANFORD NEED STATEMENT

A way to
[address problem] in
[population] to
[achieve outcome]

“Shows walking patterns”



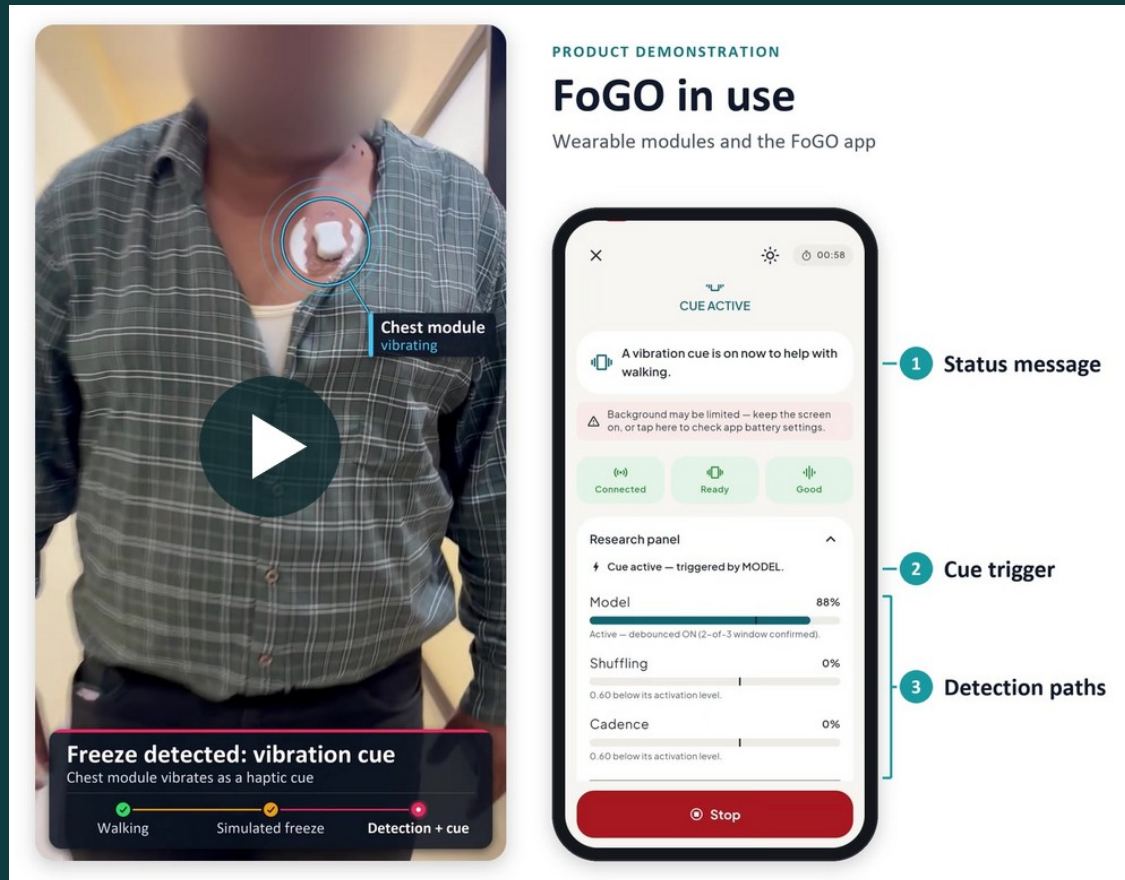
Low bar

*“Detects Parkinson’s
freezing”*



High bar: proof,
licence

What did the prototype show?



Staged demo • volunteer • simulated freeze

Seen

Stop sensed

Stop flagged as freeze

Chest vibrates

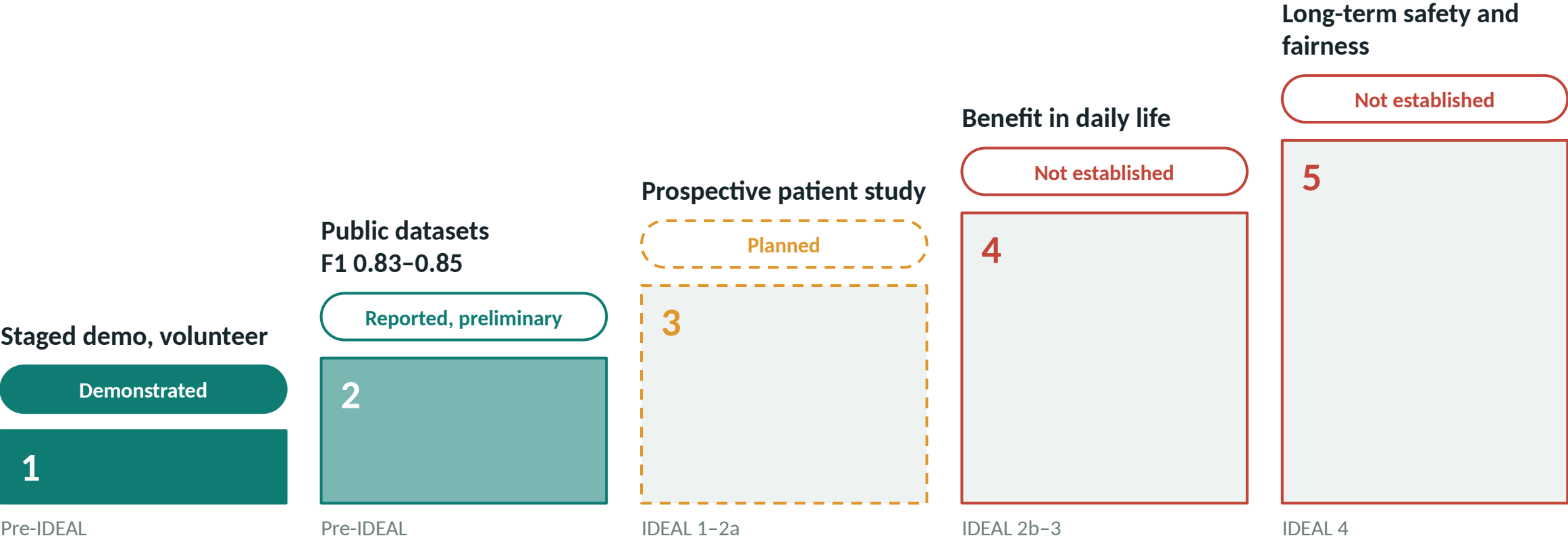
Not yet shown

Real freezes

Many users

Fewer falls

FoGO has climbed two rungs, not five



Sources: FoGO project material (supplied; project-reported); IDEAL (Lancet 2009; 2019 update); IDEAL-D (BMJ 2016)

Home pilot for Ramesh next month?



F1 0.83–0.85 on public data



Proceed



Ten users, alone at home



Proceed with conditions

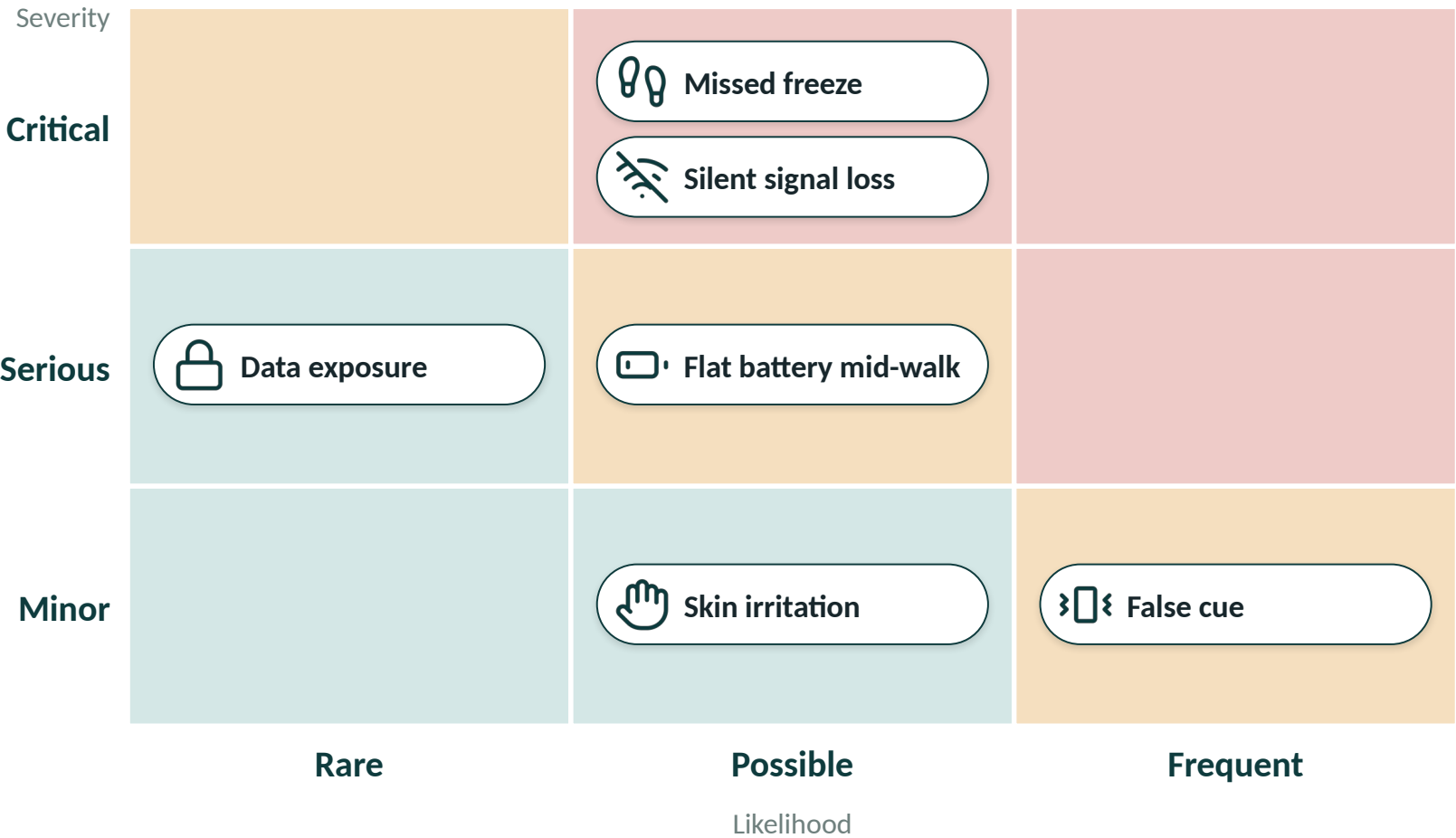


Ramesh fell twice; asks to join



Not yet

Rank failures by harm and likelihood



9-16

false cues per walking hour,
if dataset rates held

Reported: 0.15-0.27/min

Which control first?

Illustrative ranking. False-alarm rate: FoGO project material (public datasets). Approach: ISO 14971:2019; Gerke et al. (2020)

A device is also a daily routine



1 Fit the applicator

Company demonstration of use · not outcome footage

≈34 hours

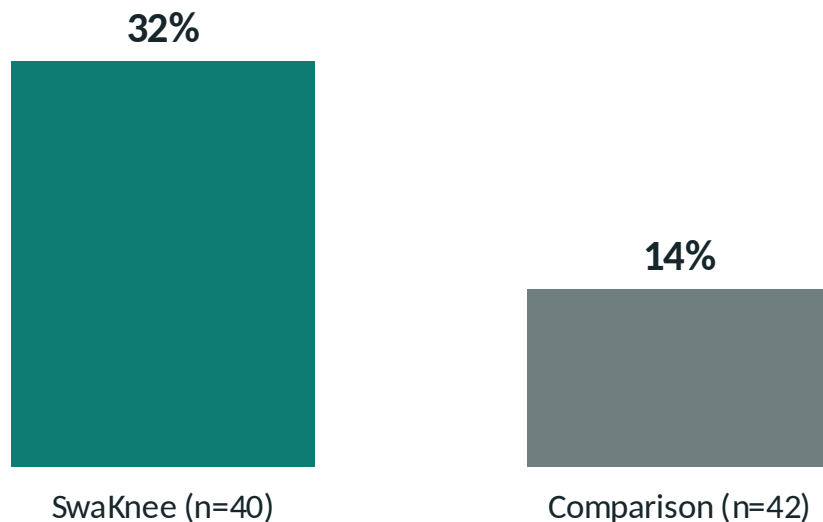
45 minutes a day for 45 days



Who helps Kamala when something goes wrong?

Which claim can this evidence carry?

Average pain reduction at day 45, company-reported (%)



Important change per patient $\approx$ 41% (Tubach 2005)

“Less pain on average, one company study”

Reported

“Clinically proven”

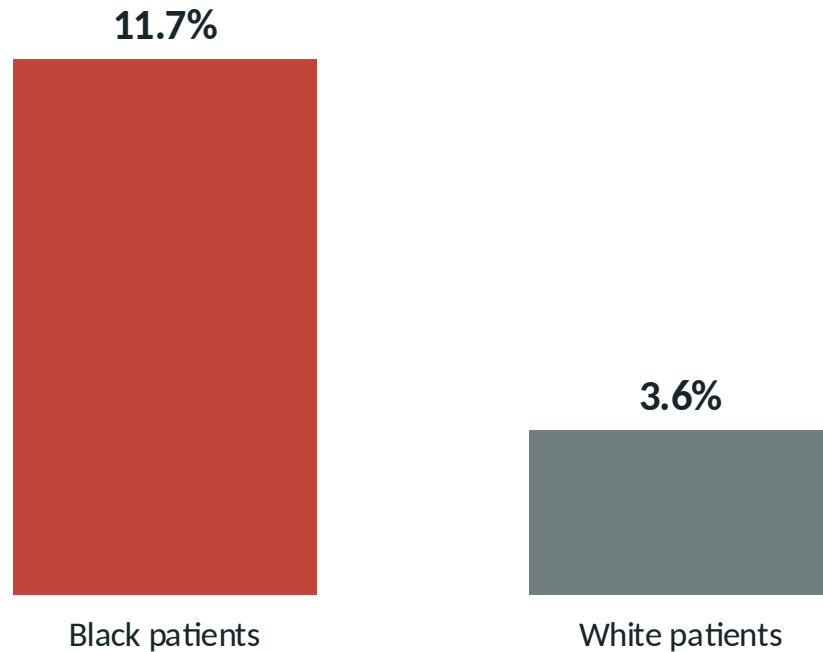
Not established

“Regrows cartilage”

Not measured

An average can hide a patient

Oximeter read 92–96% but arterial saturation was <88%



Who is missing from our data?



Ramesh: walking aids, crowded homes



Kamala: obesity, other illnesses, rural care



Report results by subgroup

His neurologist is the inventor



“Will saying no change my care?”



Recruit now



Consent form: English only



Proceed with safeguards

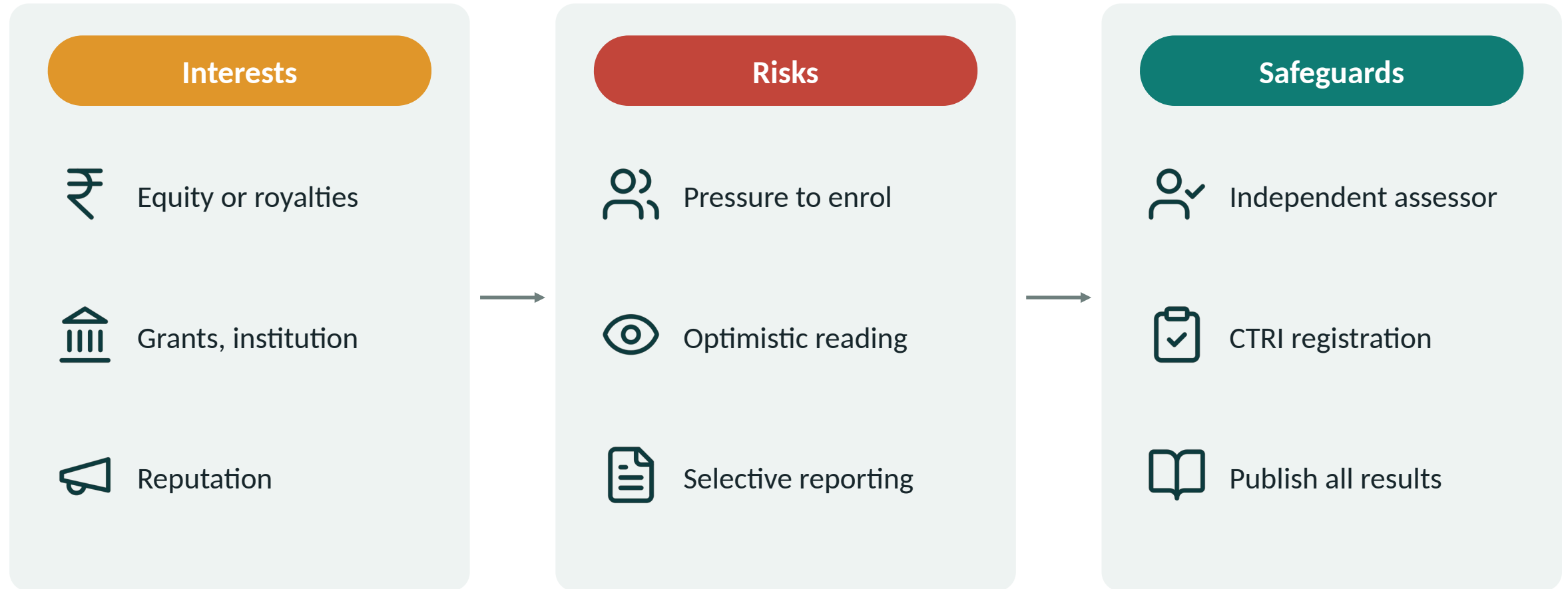


Only neurologist within 100 km

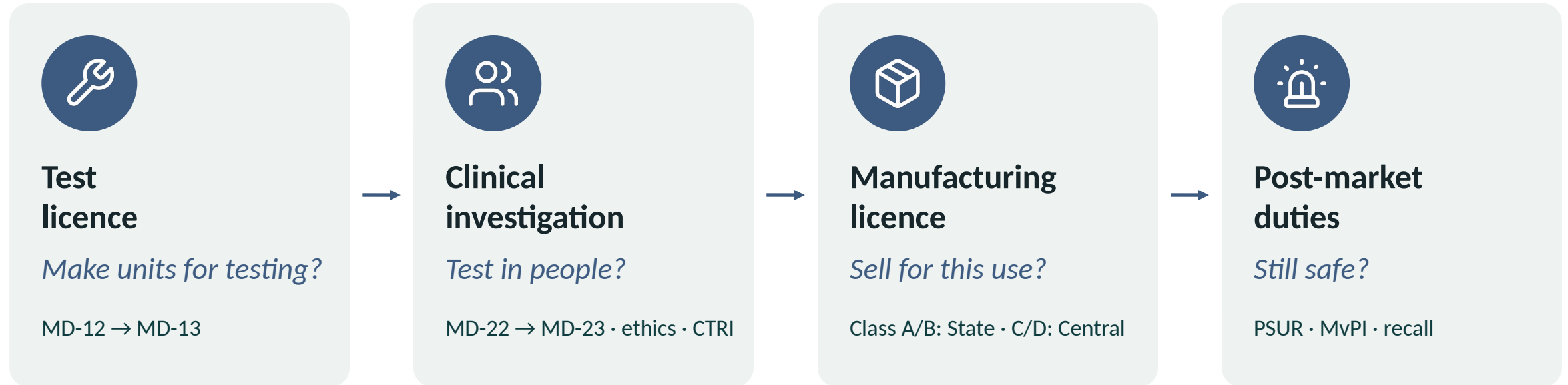


Not yet

Disclose, then manage



Each permission answers a different question



FoGO TODAY ✓ MD-13 test licence, 25 units (Aug 2026) → Next: MD-22, ethics, CTRI

Who can't use it?

	FoGO	SwaKnee
 Vision		
 Hearing		
 Thinking		
 Communication		
 Locomotion		
 Reach and stretch		
 Dexterity		
	 Low	 Medium
	 High	

INDIA ADDS

 Language

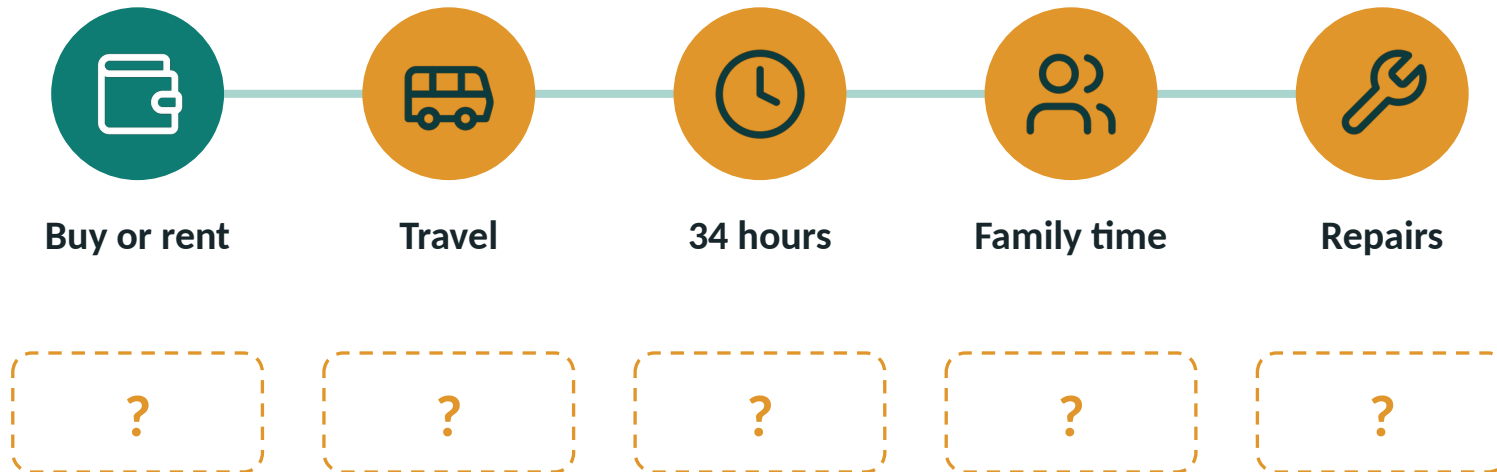
 Literacy

 Power cuts

 No smartphone

 Caregiver needed

Price is only part of the cost



Who pays each step?

NPPA PRECEDENT

91%

of home-device brands cut prices

Trade-margin cap, July 2021

Launch with this brochure?

Clinically proven!

Regrows cartilage

Doctor recommended

Avoid surgery forever

HYPOTHETICAL



Distributor insists



Cash lasts 3 months



A Launch as written

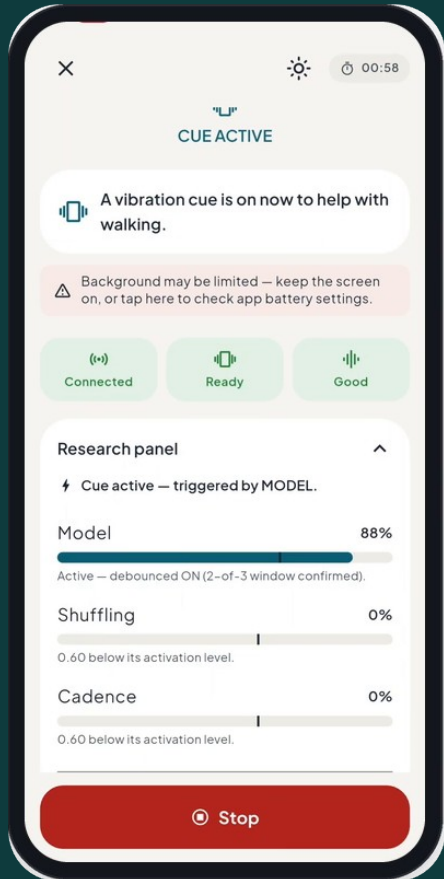


B Fix claims first



C Not yet

Push tonight's FoGO update?



Fewer false cues



May miss more freezes



Tested on stored data



Push to everyone

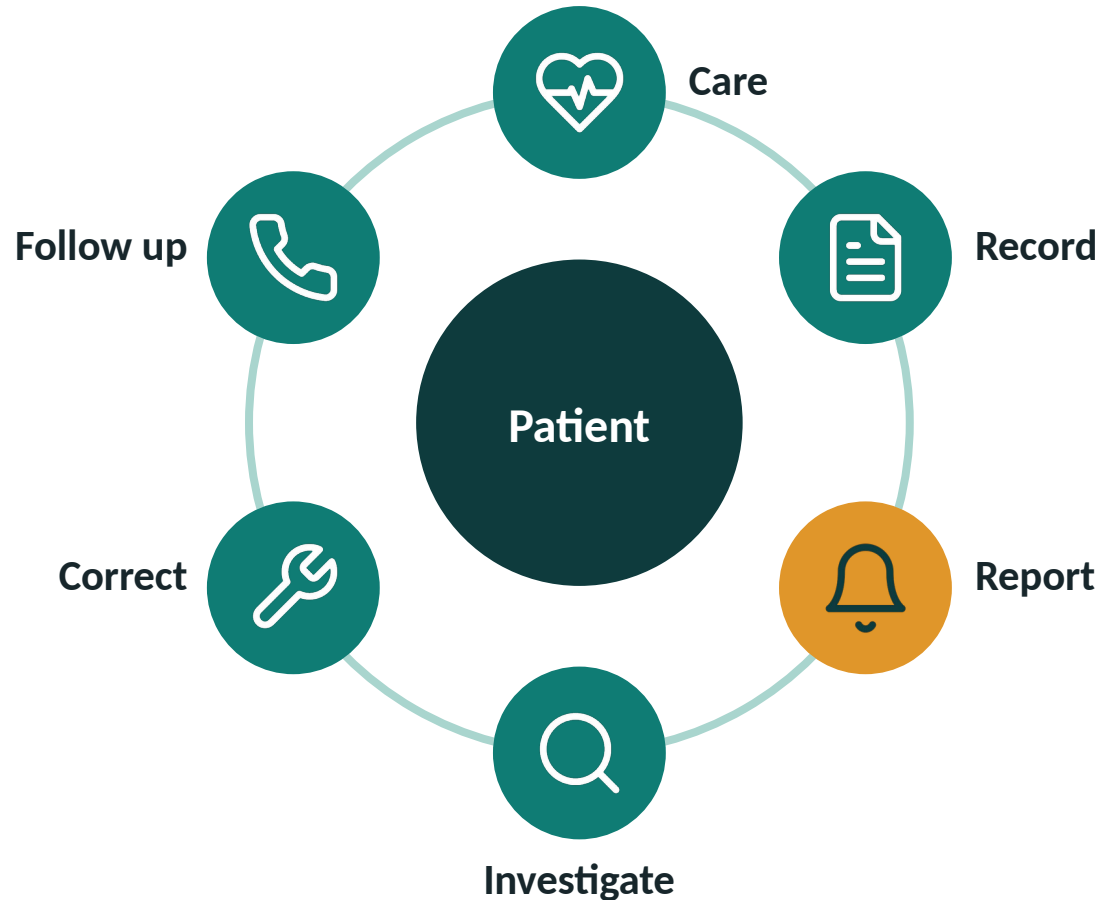


Staged release with rollback



Not yet

When harm happens, who answers?



INDIA

ASR metal-on-metal hips

≈4,700 implanted in India; recalled 2010

1,080 traced by 2018

Report harm: MvPI 1800-180-3024

Would you let a patient use it now?

Which patient?

Which use?

Which evidence?

Who is responsible?

1

Routine care

2

Research study

3

Not yet

Did your vote change? Why?

Commit to one change

Before **[next step]**, I will **[action]**.

Owner: **[name]**

I proceed only if **[condition]**.



**Write it.
Share it.
Keep it.**

Patient-impact decision checklist

Stage	Ask before proceeding	Evidence to see	Owner	Go?
Need	Is the need validated with patients and clinicians?	Need statement, observation notes	Clinical lead	☐ Yes ☐ Not yet
Evidence	Does evidence match the claim, users and version? Are failures controlled?	Protocol, subgroups, risk file, version log	Study lead	☐ Yes ☐ Not yet
People	Can people refuse freely? Are conflicts managed?	Consent in local language, COI plan	Investigator, ethics committee	☐ Yes ☐ Not yet
Permission	Which CDSCO permission covers this use?	Class, licence or permission, CTRI	Regulatory lead	☐ Yes ☐ Not yet
Access	Can users afford, use and maintain it? Do claims match evidence?	Total cost, service plan, claims review	Product and service	☐ Yes ☐ Not yet
Safety	Who records, reports and acts on harm?	Complaint log, MvPI route, update plan	Manufacturer, clinic	☐ Yes ☐ Not yet

For every “Not yet”: name the owner and the condition for proceeding.

Evidence to request before stronger claims

Question	FoGO	SwaKnee
Outcome	Freeze detection against video-annotated events; falls, confidence, walking	Pain and function with minimal important difference; structural outcomes only if claimed
Comparison	Participant-level validation; cue vs no-cue or sham cue	Randomised, blinded sham comparison; co-interventions recorded
Who	Disease stage, walking aids, homes, turning and doorways	Age, BMI, severity, comorbidities, rural access
Burden and harm	False cues, missed events, signal loss, skin, data	Adverse events, adherence to 45-minute sessions, support needs
Version	Firmware and model version for every result	Controller and applicator version for every result

What independent evidence says

Freezing-of-gait detection and cueing (FoGO)



Best-known benchmark: 10 patients, laboratory only (Daphnet, 2010)



One review: sensitivity 73–100%, specificity 67–100%, mostly in laboratories



Cueing: modest gait gains (RESCUE trial); no trial found showing fewer falls



FoGO: F1 0.83–0.85 on public data; prospective study planned (project-reported)

PEMF for knee osteoarthritis (SwaKnee)



Cochrane 2013 (all OA sites): pain probably eases ≈15/100 more than sham



Meta-analyses disagree on pain (Chen 2019, knee; Yang 2020, all OA sites)



OARSI 2019: electromagnetic therapy strongly recommended against



No human evidence found of cartilage regrowth

Teaching materials: Stanford Biodesign and Harvard (1/2)

[Stanford Biodesign, Our Process](#)

Stanford Mussallem Center for Biodesign. Our Process: Identify, Invent, Implement.

[Stanford Biodesign, Need Statements \(2022\)](#)

Stanford Biodesign. Need Statements. Student Guide to Biodesign; 2022.

[Stanford Biodesign, Principled Decision-Making \(2022\)](#)

Stanford Biodesign. Principled Decision-Making. Student Guide to Biodesign; 2022.

[Stanford Biodesign, JEDI brief \(2022\)](#)

Stanford Biodesign. Justice, Equity, Diversity and Inclusion (JEDI). Student Guide to Biodesign; 2022.

[Chao, Riskin & Krummel \(2010\)](#)

Chao KZ, Riskin DJ, Krummel TM. A patient-centered, ethical approach to medical device innovation. Virtual Mentor (now AMA J Ethics). 2010;12(2):91–95.

[Stanford GSB case OIT105 \(2011\)](#)

Burns L, Denend L, Zenios S. The Role of Physicians in Device Innovation: Critical Success Factor or Conflict of Interest? Stanford GSB case OIT105; 2011.

[Stanford Biodesign in India](#)

Stanford Biodesign. Stanford Biodesign in India (Stanford-India Biodesign, 2007–2014; now School of International Biodesign, AIIMS and IIT Delhi).

[Chaturvedi et al., BMJ Innovations \(2015\)](#)

Chaturvedi J, Logan A, Narayan G, Kuttappa S. A structured process for unmet clinical need analysis for medical device innovation in India: early experiences. BMJ Innovations. 2015;1(3):81.

[Yock, Zenios et al., Biodesign, 2nd ed. \(2015\)](#)

Yock PG, Zenios S, Makower J, Brinton TJ, Kumar UN, Watkins FTJ, et al. Biodesign: The Process of Innovating Medical Technologies. 2nd ed. Cambridge University Press; 2015 (Stanford Biodesign content).

[Stanford Biodesign, Health Equity](#)

Stanford Biodesign. Health Equity programme and resources.

[HMS Center for Bioethics](#)

Harvard Medical School Center for Bioethics. Fellowship in Bioethics: curriculum.

[MRCT Center, Diversity guidance \(2020\)](#)

MRCT Center of Brigham and Women's Hospital and Harvard. Achieving Diversity, Inclusion, and Equity in Clinical Research: Guidance Document and Toolkit; 2020.

Teaching materials: Stanford Biodesign and Harvard (2/2)

[MRCT Center, Accessibility by Design \(2023\)](#)

MRCT Center. Accessibility by Design in Clinical Research Toolkit; 2023.

[MRCT Center, Clinical Research Glossary](#)

MRCT Center. Clinical Research Glossary (plain-language definitions; v3.0, 2025).

[MRCT Center, post-trial access for devices \(2025\)](#)

MRCT Center Post-trial Continued Access Task Force. Post-trial, Continued Access Responsibilities to Investigational Significant-Risk Devices: Framework; 2025.

[MRCT Center, Return of Aggregate Results \(2017\)](#)

MRCT Center. Return of Aggregate Results to Participants: Guidance Document and Toolkit, v3.1; 2017.

[Petrie-Flom, Diagnosing in the Home](#)

Petrie-Flom Center, Harvard Law School. Diagnosing in the Home: the ethical, legal and regulatory challenges of digital home health (2021–2024).

[Gerke, Babic, Evgeniou & Cohen \(2020\)](#)

Gerke S, Babic B, Evgeniou T, Cohen IG. The need for a system view to regulate artificial intelligence/machine learning-based software as medical device. npj Digit Med. 2020.

[Simon, Shachar & Cohen, JAMA \(2022\)](#)

Simon DA, Shachar C, Cohen IG. Unsettled liability issues for 'prediagnostic' wearables and health-related products. JAMA. 2022;328:1391.

[Cohen, Gerke & Kramer, Milbank Q \(2020\)](#)

Cohen IG, Gerke S, Kramer DB. Ethical and legal implications of remote monitoring of medical devices. Milbank Q. 2020;98(4):1257–1289.

[HMS conflict-of-interest policy \(2010\)](#)

Harvard Medical School. Policy on Conflicts of Interest and Commitment (2010 revision).

[Harvard Catalyst Regulatory Foundations](#)

Harvard Catalyst. Regulatory Foundations, Ethics, and Law Program.

[HBS case, Aravind Eye Hospital \(1993\) • no public link](#)

Rangan VK. The Aravind Eye Hospital, Madurai, India: In Service for Sight. Harvard Business School case 593-098; 1993.

[Kramer et al., PLoS Med \(2013\)](#)

Kramer DB, Tan YT, Sato C, Kesselheim AS. Postmarket surveillance of medical devices: a comparison of strategies in the US, EU, Japan, and China. PLoS Med. 2013.

Teaching materials: Cambridge, and evaluation frameworks

Cambridge EDC, Inclusive Design Toolkit

University of Cambridge Engineering Design Centre. Inclusive Design Toolkit and Exclusion Calculator (Clarkson, Waller, Coleman).

Engineering Better Care (2017)

Royal Academy of Engineering, Academy of Medical Sciences, Royal College of Physicians. Engineering Better Care: a systems approach to health and care design and continuous improvement; 2017 (working group chaired by Prof P. John Clarkson, University of Cambridge).

Cambridge CEBC, Improving Improvement toolkit

University of Cambridge Centre for Engineering Better Care. Improving Improvement: a toolkit for Engineering Better Care.

Cambridge Engineering module 4G9

University of Cambridge Department of Engineering. Part IIB module 4G9: Biomedical Engineering / Biomedical Innovation (syllabus).

Cambridge Institute for Biomedical Innovation (2026)

University of Cambridge. New Cambridge institute will support high-value medical device translation; January 2026.

Cambridge Judge, CCHLE

Cambridge Judge Business School. Centre for Health Leadership and Enterprise.

University of Cambridge COI policy

University of Cambridge. Conflict of Interest Policy.

IE Cambridge, How to Guide: Spinouts (2024)

University of Cambridge Innovation and Entrepreneurship. How to Guide 3: Spinouts; 2024.

Clarkson, Design Science (2018)

Clarkson PJ. What has engineering design to say about healthcare improvement? Design Science. 2018;4:e17.

McCulloch et al., Lancet (2009), IDEAL

McCulloch P, Altman DG, Campbell WB, et al. No surgical innovation without evaluation: the IDEAL recommendations. Lancet. 2009;374:1105–1112 (IDEAL Collaboration; 2019 update adds Pre-IDEAL).

Sedrakyan et al., BMJ (2016), IDEAL-D

Sedrakyan A, Campbell B, Merino JG, Kuntz R, Hirst A, McCulloch P. IDEAL-D: a rational framework for evaluating and regulating the use of medical devices. BMJ. 2016;353:i2372.

Kellmeyer et al., Camb Q Healthc Ethics (2016)

Kellmeyer P, Cochrane T, Müller O, et al. The effects of closed-loop medical devices on the autonomy and accountability of persons and systems. Camb Q Healthc Ethics. 2016;25(4):623–633.

India: regulation, ethics, pricing and safety (1/2)

Medical Devices Rules, 2017

Ministry of Health and Family Welfare. Medical Devices Rules, 2017 (as amended), consolidated text, CDSCO.

CDSCO Form MD-12 (test licence application)

CDSCO. Form MD-12: application for licence to manufacture medical devices for clinical investigation, test, evaluation, examination, demonstration or training (licence in Form MD-13).

CDSCO Form MD-22 (clinical investigation)

CDSCO. Form MD-22: application for permission to conduct clinical investigation (permission in Form MD-23).

CDSCO, Regulatory pathway under MDR-2017

CDSCO. Regulatory pathway under the Medical Devices Rules, 2017 (overview).

CDSCO, Medical Device Software guidance (21 Jul 2026)

CDSCO. Guidance Document on Medical Device Software under the Medical Devices Rules, 2017 (CDSCO/MD/GD/MDSW/01/2026); 21 July 2026.

CDSCO draft software guidance (21 Oct 2025)

CDSCO. Draft Guidance Document on Medical Device Software; 21 October 2025.

New Drugs and Clinical Trials Rules, 2019

Ministry of Health and Family Welfare. New Drugs and Clinical Trials Rules, 2019 (ethics committee registration: Rules 7–8 with CDSCO; Rules 16–17 with DHR).

ICMR National Ethical Guidelines (2017)

Indian Council of Medical Research. National Ethical Guidelines for Biomedical and Health Research Involving Human Participants. New Delhi: ICMR; 2017 (Section 5 informed consent; Section 6 vulnerability; Section 7.7 device trials).

CDSCO draft guidance on post-market surveillance (2018)

CDSCO. Draft Guidance on Post Market Surveillance of Medical Devices; 2018.

IPC, Materiovigilance Programme of India

Indian Pharmacopoeia Commission. Materiovigilance Programme of India (MvPI): About us. Launched 6 July 2015; helpline 1800-180-3024; mvpi-ipc@gov.in.

MvPI Medical Device Adverse Event Reporting Form

Indian Pharmacopoeia Commission. Medical Device Adverse Event Reporting Form, version 1.2 (open to manufacturers, importers, distributors, healthcare professionals and patients).

DPDP Rules, 2025

Ministry of Electronics and Information Technology. Digital Personal Data Protection Rules, 2025 (notified November 2025; phased commencement).

Uniform Code for Marketing Practices in Medical Devices (2024, amended 2026)

Department of Pharmaceuticals. Uniform Code for Marketing Practices in Medical Devices, 2024 (circular of 6 September 2024; as amended on 30 April 2026). Voluntary code administered through industry associations.

India: regulation, ethics, pricing and safety (2/2)

IMC Professional Conduct Regulations 2002, clause 6.8

Medical Council of India. Indian Medical Council (Professional Conduct, Etiquette and Ethics) Regulations, 2002, clause 6.8 (inserted 2009); operative after the NMC 2023 regulations were held in abeyance on 23 August 2023.

NMC 2023 conduct regulations in abeyance

The Wire. NMC puts new guidelines on doctors' ethics in abeyance; August 2023.

Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954

Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 (Schedule includes 'rheumatism').

ASCI Code, Chapter I.1 (CCC press release)

Advertising Standards Council of India. Consumer Complaints Council recommendations press release (June 2022): health advertisements assessed under ASCI Code Chapter I.1 (truthful, substantiated claims) and III.4.

PIB/NPPA: coronary stent ceiling prices (13 Feb 2017)

Press Information Bureau. NPPA ceiling prices for coronary stents: Rs 7,260 (bare-metal) and Rs 29,600 (drug-eluting); 13 February 2017.

PIB/NPPA: knee implant ceiling prices (16 Aug 2017)

Press Information Bureau. Government fixes ceiling prices of knee implants (cobalt-chromium primary knee: average MRP Rs 1,58,324 to Rs 54,720, 65%; special metals 69%); 16 August 2017.

Knee implant cap extended to 15 Nov 2026

Medical Dialogues. NPPA continues price cap on orthopaedic knee replacement implants for another year, till 15 November 2026.

PIB/NPPA: trade-margin cap on five home-use devices (13 Jul 2021)

Press Information Bureau. Government caps trade margin up to 70% on price-to-distributor for pulse oximeters, BP monitors, nebulisers, digital thermometers and glucometers; 13 July 2021 (91% of 684 brands cut MRP, by up to 88%).

TCTMD: manufacturers' responses to India's stent cap

TCTMD. Stents as essential medicine: India's cap on stent prices could have ripples around the globe; 2017 (withdrawal applications refused).

ASR hip implants: MoHFW expert committee (2018)

Business Today. Govt to ask Johnson & Johnson to compensate for faulty hip implants (Agarwal committee report, 19 Feb 2018: about 4,700 ASR surgeries in India 2004–2010; 1,080 patients traced); 25 August 2018.

PIB: compensation formula for ASR hip implant cases (29 Nov 2018)

Press Information Bureau. Health Ministry approves compensation formula for hip implant cases (Central Expert Committee; base amount Rs 20 lakh adjusted for disability and age); 29 November 2018.

Pal et al., Indian J Orthop (2016)

Pal CP, Singh P, Chaturvedi S, Pruthi KK, Vij A. Epidemiology of knee osteoarthritis in India and related factors. Indian J Orthop. 2016;50(5):518–522 (28.7% in a five-site community survey).

India State-Level Disease Burden Initiative, Lancet Glob Health (2021)

India State-Level Disease Burden Initiative Neurological Disorders Collaborators. The burden of neurological disorders across the states of India: GBD 1990–2019. Lancet Glob Health. 2021.

Ethics and international comparison (1/2)

WMA Declaration of Helsinki (2024)

World Medical Association. Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Participants. 75th General Assembly, Helsinki, October 2024 (para 27: dependent relationship; para 34: post-trial provisions).

CIOMS International Ethical Guidelines (2016)

CIOMS with WHO. International Ethical Guidelines for Health-related Research Involving Humans. 4th ed. Geneva; 2016 (Guideline 9 consent; Guideline 25 conflicts of interest).

Institute of Medicine (2009). Conflict of Interest

Institute of Medicine; Lo B, Field MJ, eds. Conflict of Interest in Medical Research, Education, and Practice. National Academies Press; 2009.

Lundh et al., Cochrane (2017)

Lundh A, Lexchin J, Mintzes B, Schroll JB, Bero L. Industry sponsorship and research outcome. Cochrane Database Syst Rev. 2017;(2):MR000033 (drug and device studies: favourable conclusions RR 1.34, 95% CI 1.19–1.51).

Appelbaum et al., Int J Law Psychiatry (1982)

Appelbaum PS, Roth LH, Lidz C. The therapeutic misconception: informed consent in psychiatric research. Int J Law Psychiatry. 1982;5(3–4):319–329.

Appelbaum et al., Hastings Cent Rep (1987)

Appelbaum PS, Roth LH, Lidz CW, Benson P, Winslade W. False hopes and best data: consent to research and the therapeutic misconception. Hastings Cent Rep. 1987;17(2):20–24.

Johnson & Rogers, J Med Ethics (2012)

Johnson J, Rogers W. Innovative surgery: the ethical challenges. J Med Ethics. 2012;38(1):9–12.

Wexler & Largent, Nat Commun (2023)

Wexler A, Largent E. Ethical considerations for researchers developing and testing minimal-risk devices. Nat Commun. 2023;14.

GMC, recordings of patients (2011, updated 2024)

General Medical Council (UK). Making and using visual and audio recordings of patients (effective 2011; updated December 2024). International comparator.

US Common Rule, key information first (45 CFR 46.116)

US HHS. Protection of Human Subjects, 45 CFR 46.116(a)(5)(i): consent must begin with a concise presentation of key information. International comparator.

US FDA, significant vs non-significant risk device studies

US FDA. Information Sheet: Significant Risk and Nonsignificant Risk Medical Device Studies (21 CFR 812).

Ethics and international comparison (2/2)

US FDA, PCCP guidance for AI-enabled device software (Dec 2024)

US FDA. Marketing Submission Recommendations for a Predetermined Change Control Plan for Artificial Intelligence-Enabled Device Software Functions. Final guidance, December 2024 (revised version reported August 2025).

US FDA, pulse oximeter draft guidance (Jan 2025)

US FDA. Pulse Oximeters for Medical Purposes: Non-Clinical and Clinical Performance Testing, Labeling, and Premarket Submission Recommendations. Draft guidance, 7 January 2025 (diverse skin-tone testing, Monk Skin Tone scale).

EU MDR 2017/745 (as amended by 2023/607)

Regulation (EU) 2017/745 on medical devices, Art. 61 and Annex XIV Part B (post-market clinical follow-up); transition extended to 2027/2028 by Regulation (EU) 2023/607.

European Commission targeted MDR/IVDR revision proposal (Dec 2025)

European Commission. Proposal for a targeted revision of the medical device regulations, COM(2025) 1023 final, 16 December 2025 (under negotiation).

Cumberlege Review, First Do No Harm (2020)

Independent Medicines and Medical Devices Safety Review (chair Baroness Cumberlege). First Do No Harm; 8 July 2020.

CMS Open Payments (US Sunshine Act)

Centers for Medicare & Medicaid Services. Open Payments: public reporting of industry payments to physicians and teaching hospitals.

ISO 14971:2019, risk management

ISO 14971:2019. Medical devices — Application of risk management to medical devices.

ISO 14155:2026, device clinical investigation GCP

ISO 14155:2026. Clinical investigation of medical devices for human subjects — Good clinical practice (supersedes ISO 14155:2020).

IEC 62366-1:2015+A1:2020, usability engineering

IEC 62366-1:2015+AMD1:2020. Medical devices — Part 1: Application of usability engineering to medical devices.

Kadakia et al., Ann Intern Med (2024)

Kadakia KT, et al. Class I recalls of cardiovascular devices between 2013 and 2022: a cross-sectional analysis. Ann Intern Med. 2024;177(11) (19.1% had premarket clinical testing).

Case evidence, media and status labels (1/2)

[Sjoding et al., NEJM \(2020\)](#)

Sjoding MW, Dickson RP, Iwashyna TJ, Gay SE, Valley TS. Racial bias in pulse oximetry measurement. *N Engl J Med*. 2020;383(25):2477–2478. doi:10.1056/NEJMc2029240.

[Bächlin et al. \(2010\), Daphnet dataset](#)

Bächlin M, Plotnik M, Roggen D, et al. Wearable assistant for Parkinson's disease patients with the freezing of gait symptom. *IEEE Trans Inf Technol Biomed*. 2010 (10 patients in the laboratory; 8 froze; 237 video-labelled events).

[Pardoel et al., Sensors \(2019\)](#)

Pardoel S, Kofman J, Nantel J, Lemaire ED. Wearable-sensor-based detection and prediction of freezing of gait in Parkinson's disease: a review. *Sensors*. 2019;19(23):5141.

[Silva de Lima et al., J Neurol \(2017\)](#)

Silva de Lima AL, Evers LJW, Hahn T, et al. Freezing of gait and fall detection in Parkinson's disease using wearable sensors: a systematic review. *J Neurol*. 2017 (sensitivity 73–100%, specificity 67–100%).

[Ginis et al., Ann Phys Rehabil Med \(2018\)](#)

Ginis P, Nackaerts E, Nieuwboer A, Heremans E. Cueing for people with Parkinson's disease with freezing of gait: a narrative review. *Ann Phys Rehabil Med*. 2018;61(6):407–413.

[Nieuwboer et al., RESCUE trial \(2007\)](#)

Nieuwboer A, et al. Cueing training in the home improves gait-related mobility in Parkinson's disease: the RESCUE trial. *J Neurol Neurosurg Psychiatry*. 2007;78(2):134 (n=153; small gait gains and lower freezing severity in freezers; fall counts not measured).

[Li et al., Cochrane \(2013\), electromagnetic fields for OA](#)

Li S, Yu B, Zhou D, et al. Electromagnetic fields for treating osteoarthritis. *Cochrane Database Syst Rev*. 2013;(12):CD003523 (9 trials, 636 adults: pain probably improves about 15/100 more than sham; function uncertain).

[Chen et al., J Rehabil Med \(2019\)](#)

Chen L, Duan X, Xing F, et al. Effects of pulsed electromagnetic field therapy on pain, stiffness and physical function in patients with knee osteoarthritis: a systematic review and meta-analysis. *J Rehabil Med*. 2019;51(11):821–827 (no pain advantage; function improved).

[Yang et al., Phys Ther \(2020\)](#)

Yang X, He H, Ye W, Perry TA, He C. Effects of pulsed electromagnetic field therapy on pain, stiffness, physical function, and quality of life in patients with osteoarthritis: a systematic review and meta-analysis. *Phys Ther*. 2020;100(7):1118–1131.

Case evidence, media and status labels (2/2)

Markovic et al., Wien Klin Wochenschr (2022)

Markovic L, Wagner B, Crevenna R. Effects of pulsed electromagnetic field therapy on outcomes associated with osteoarthritis: a systematic review of systematic reviews. Wien Klin Wochenschr. 2022 (heterogeneous schemes; no sufficient proof of efficacy).

OARSI guideline (2019)

Bannuru RR, Osani MC, Vaysbrot EE, et al. OARSI guidelines for the non-surgical management of knee, hip, and polyarticular osteoarthritis. Osteoarthritis Cartilage. 2019 (electromagnetic therapy strongly recommended against).

NICE NG226 (2022)

NICE. Osteoarthritis in over 16s: diagnosis and management (NG226); October 2022 (do not offer listed electrotherapies: insufficient evidence of benefit).

Tubach et al., Ann Rheum Dis (2005)

Tubach F, Ravaud P, Baron G, et al. Evaluation of clinically relevant changes in patient reported outcomes in knee and hip osteoarthritis: the minimal clinically important improvement. Ann Rheum Dis. 2005;64(1):29–33 (knee OA pain MCII –19.9 mm, –40.8%).

FoGO BIRAC BIG project presentation, supplied

Ahilaya Biomedicals. FoGO BIRAC BIG presentation (supplied, 2026): alpha prototype; public-dataset F1 0.83–0.85; MD-13 test licence (25 units, Aug 2026); AIIMS Bhubaneswar proof-of-concept planned. Project-reported.

SwaKnee leaflet, supplied

Swa knee_Leaflet.pdf. Supplied product material: 45-minute daily sessions over 45 days (illustration of time burden, not a prescription).

Swayogya evidence page (company-reported)

Swayogya Rehab Solutions. SwaKnee evidence page: company-reported comparative 45-day study, device n=40 vs comparison n=42, ≈32% vs ≈14% average VAS pain reduction from baseline. Rounded figures as transcribed in the supplied original deck; re-check the live page before presenting.

SwaKnee photographs and step videos

Swayogya website assets (ivikasjha/swayogya_website): product photographs and three 'How to use' step videos, edited into a 20 s silent clip with step captions; no faces shown.

SwaKnee explainer video (company, optional)

Swayogya Rehab Solutions. SwaKnee explainer video on YouTube (company material; optional, longer than the embedded clip).